

Homework 1: Introduction to Text Mining and Natural Language Processing

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February 5, 2025

Introduction

In this research project we analyze the impact of the Mobile World Congress (MWC) of Barcelona in 2025¹ on the city's rental prices of Booking. To do so, we rely on data for the full population of accommodations²:

- For the first 3 weeks of each month from February to June 2025, and
- For Barcelona (**treatment** group) and Madrid (**control** group).

Taking this into account, we have used the week from the 9th to the 16 of February as the **pre-treatment period** and the week from the 1st to the 8th of March as the **treatment period**.

Part 1: Scraping

As introduced above, we have scraped data for all of the available accommodations in Barcelona and Madrid for the first 3 weeks from February to June 2025. More specifically, we have extracted the following information for each accommodation, which we considered as potentially relevant for explaining and/or controlling the effect on prices of the MWC:

1. Name of the accommodation.
2. Price (in euros). The value reflects the price for renting the accommodation for a week.
3. Rating.
4. Number of reviews.
5. Quality of the accommodation over 5 (e.g., number of stars in the case of the hotels).
6. Neighborhood where the accommodation is located.
7. The short description that is shown in the search results, which commonly describes basic information about the room and whether breakfast is included in the price.
8. The URL, which we have used to extract the "long" description.
9. The "long" description of each accommodation, which provides a brief summary to the user of the closeness of the accommodation to major touristic attractions, the facilities offered in the accommodation, etc.

The code that has scraped these data has been developed using the *Selenium* and *requests* Python packages. It is completely automated. In short:

- It uses *Selenium* and loops over the specified places and months, loads all of the results for each week and location and extracts the data of the points 1 to 8 above.

¹Which lasts from the 3rd to the 6th of March.

²Retrieved on the 29th of January 2025.

- After iterating over all the months and locations that have been set, it uses *requests* to extract all of the descriptions for all of the unique accommodations.
- The results are saved progressively, to avoid data loss during the process.

For more information on the specific code, refer to the attached Python scripts.

Part 2: Text Analysis

As a result of the scraping process, this analysis takes two main sets of files as input:

- **Hotel Listings:** Files containing hotel availability and pricing information for two time periods (February 9-16 and March 1-8, 2025).
- **Hotel Descriptions:** A dataset with text descriptions of the hotels, serving as the main source for text analysis.

The analysis follows these steps:

1. Load the hotel listings and descriptions.
2. Filter the dataset to keep only hotels that appear in both time periods for consistency.

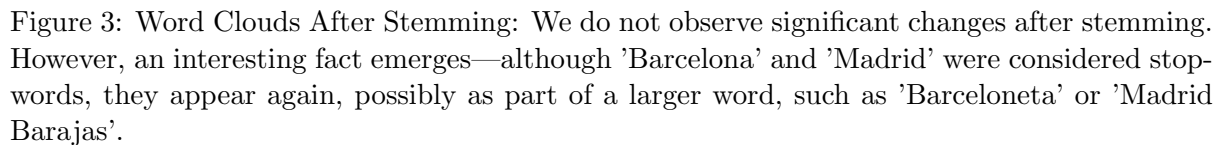
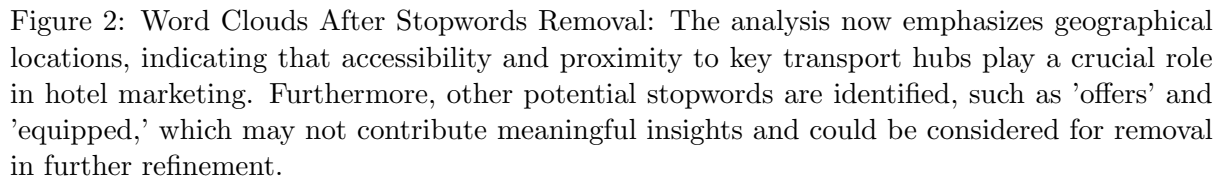
To prepare the text for analysis, the following cleaning steps are applied:

- Convert text to lowercase.
- Remove punctuation, common stopwords in English, and custom stopwords derived from an analysis of earlier word cloud generations (e.g., “hotel,” “barcelona”, “madrid”).
- Apply **stemming** for standarization.

Word clouds are generated before and after preprocessing to highlight changes in word frequency. This visualization helps identify commonly used terms in hotel descriptions for Madrid and Barcelona.



Figure 1: Word Clouds Before Pre-processing: At first glance, we notice the presence of some words, such as 'Barcelona', 'Madrid', 'hotel', and 'km' that do not contribute to the analysis. Therefore, we decided to include them as custom stopwords.



- Identifies words that are important in each city's hotel descriptions.
- Highlights terms unique to Madrid or Barcelona beyond just frequently occurring words.



Figure 4: In the TF-IDF Word Cloud, some terms (e.g., "plaza," "district") still appear prominently, reinforcing the idea that geographical names hold significant contextual relevance in hotel descriptions. It's quite curious that "beach" appears to have a higher weighting in the analysis compared to "Sagrada Familia," which is arguably Barcelona's most iconic landmark. "Rambla" didn't appear significantly in the initial word cloud analysis, but now it is a more prominent term in the TF-IDF word cloud.

The final dataset and TF-IDF results are exported in CSV format for further analysis.

Part 3: Difference-in-Differences

Differences-in-Differences Model Setup

Objective: We want to understand the effect on hotel prices of the event *Mobile World Congress* in Barcelona from **2025-03-01** to **2025-03-08**.

Available Data

We have data for hotel prices, quality, rating, and reviews for two cities: **Barcelona** (treatment group) and **Madrid** (control group).

Time Period Considered

The analysis covers three distinct time periods across six datasets (three for Barcelona and three for Madrid):

1. **Pre-Treatment:** 2025-02-09 to 2025-02-16 (Barcelona and Madrid)
2. **Treatment/Event:** 2025-03-01 to 2025-03-08 (Barcelona and Madrid)
3. **Post-Treatment:** 2025-04-09 to 2025-04-16 (Barcelona and Madrid)

Analysis Approach

To implement the Differences-in-Differences (DiD) model:

- We check for **parallel trends** in the pre-treatment period. We see common trends pre-event and post-event for Barcelona and Madrid.
- A simple difference in difference table using average price price effect analysis also shows clear impact of Mobile world congress increased hotel prices in Barcelona relative to Madrid. However, this effect is inflated because the table doesn't account for control variables.
- We also examine trends in the post-treatment period for additional validation.

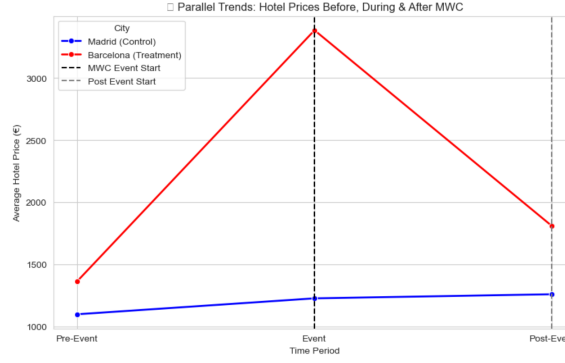


Figure 5: Parallel trends

****Difference-in-Differences (DID) Table: Impact of Mobile World Congress****

Difference-in-Differences (DID) Table: Impact of Mobile World Congress

	Before Event (€)	During Event (€)	Price Change (Δ) (€)
Madrid (Control Group)	1,181.40	1,228.20	46.80
Barcelona (Treatment Group)	1,597.68	3,384.58	1,786.90
DID Estimate (MWC Impact)	nan	nan	1,740.10

Figure 6: Difference in Difference Table

Difference-in-Differences (DID) Model

- We then ran the **DID regression analysis** during the treatment period.

The DID model for Mobile World Congress Impact on Hotel Prices, which we implement is given as:

Part 3a - Regression Equation

$$\text{price_euros}_{it} = \beta_0 + \beta_1 \text{Barcelona}_i + \beta_2 \text{Event}_t + \beta_3 (\text{Barcelona}_i \times \text{Event}_t) + \sum_k \delta_k X_{kit} + \epsilon_{it} \quad (1)$$

Variable Definitions

- price_euros_{it} = Hotel price for hotel i at time t (dependent variable)
- Barcelona_i = 1 if the hotel is in Barcelona, 0 otherwise (treatment group indicator)
- Event_t = 1 for event period (2025-03-01 to 2025-03-08), 0 for pre-event period (2025-02-08 to 2025-02-16)
- $\text{Barcelona}_i \times \text{Event}_t$ = Interaction term capturing the causal effect of MWC on prices
- X_{kit} = Control variables: rating (rating of the hotel), quality_over_5 (rating over 5), quality_type (type of the hotel), num_reviews (number of reviews), neighborhood (neighborhood where the hotel is located)
- Text features: max_tfidf_hotel (Max tfidf score of the hotel) and unique_tfidf_word_cnt (Unique tfidf word count for the hotel)
- ϵ_{it} = Error term

Interpretation of Coefficients

- β_0 : Baseline hotel prices in Madrid during pre-event

- β_1 : Difference in baseline prices between Barcelona and Madrid (before the event)
 - β_2 : Change in Madrid's hotel prices due to the event period (event-driven price change in control group)
 - β_3 (**Key Coefficient**): Causal effect of MWC on Barcelona hotel prices (DID effect)
- Part 3a (Cont.) - We need a 2nd city with a similar hotel trend as Barcelona to ensure spike in hotel prices is because of the event we are studying on the assumption of parallel trends between prices in both cities. The only major difference in hypothetical situation between the two cities should be the event ie Mobile World congress in this case.
 - Part 3b - We created multiple text features basis the descriptions given by the hotel, but used max_tfidf_hotel (Max tfidf score of the hotel) and unique_tfidf_word_cnt (Unique tfidf word count for the hotel) in the final model , which are coming significant.
 - Note the DID effect varies for the regression (1787) relative to the averages computed from the table (1700) because of the following reasons -
 - Controls for Confounding Factors: The regression model adjusts for quality type, rating, number of reviews, and neighborhood differences.
 - More Stable Estimate: Unlike the raw mean difference, OLS regression balances price variations by minimizing error terms.
 - Regression reduces Impact of Outliers: High-price hotels (luxury properties) in Barcelona don't disproportionately affect results.

Regression for DID Model

Table 1: OLS Regression Results

Variable	Coef.	Std. Err.	t	P> t	Lower Bound	Upper Bound
Intercept	-1477.9064	363.412	-4.067	0.000	-2190.599	-765.214
C(quality_type)[T.stars]	-54.2357	78.651	-0.690	0.491	-208.479	100.007
C(neighborhood)[T.Barajas, Madrid]	117.1189	311.963	0.375	0.707	-494.677	728.915
C(neighborhood)[T.Eixample, Barcelona]	322.2073	123.779	2.603	0.009	79.463	564.952
C(neighborhood)[T.Sants-Montjuïc, Barcelona]	1366.3950	175.655	7.779	0.000	1021.916	1710.874
Barcelona	173.2575	206.627	0.839	0.402	-231.963	578.478
Event	33.7721	79.243	0.426	0.670	-121.633	189.178
Barcelona:Event	1787.6765	122.687	14.571	0.000	1547.074	2028.279
rating	142.2077	38.704	3.674	0.000	66.306	218.110
quality_over_5	372.8483	41.902	8.898	0.000	290.675	455.022
num_reviews	-0.0739	0.018	-4.184	0.000	-0.108	-0.039
max_tfidf_hotel	26.2391	11.549	2.272	0.023	3.590	48.888
unique_tfidf_word_cnt	9.0394	5.859	1.543	0.123	-2.452	20.531

Notes: Standard Errors assume that the covariance matrix of the errors is correctly specified. The smallest eigenvalue is $1e-22$, which might indicate strong multicollinearity problems.

Estimated Difference-in-Differences (DID) Effect: 1787.6765

Table 2: Regression Statistics

R-squared	0.327
Adj. R-squared	0.315
F-statistic	27.06
Prob (F-statistic)	8.80e-149
Observations	2098
AIC	36220
BIC	36440
Durbin-Watson	1.737

Part 3c - Decomposing the treatment effect by hotel quality.

1. Deriving Hotel Quality Categories from Text Descriptions

- **Topic Modeling (NMF/LDA):** Classify hotels into luxury, budget, business, family-friendly segments.
- **Sentiment Analysis:** Higher positive sentiment suggests better quality.
- **TF-IDF Keyword Presence:** Hotels with high frequency of words like “luxury,” “premium,” “spa” may be high-end.
- **Lexical Complexity:** More complex and descriptive text may indicate higher-end hotels.

2. Regression Model: DID with Interaction Terms

We use a Difference-in-Differences (DID) regression with interaction terms to estimate how the treatment effect varies by hotel quality.

Regression Equation

Let:

- Y_{ict} = Hotel price for hotel i in city c at time t
- Post_t = 1 if post-event, 0 otherwise
- Barcelona_c = 1 if city is Barcelona, 0 if control
- Quality_i = Hotel quality category (e.g., luxury, budget, mid-tier)
- X_i = Other hotel characteristics (e.g., rating, amenities, TF-IDF features)
- ϵ_{ict} = Error term

The regression model is specified as:

$$Y_{ict} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Barcelona}_c + \beta_3 (\text{Post}_t \times \text{Barcelona}_c) + \beta_4 \text{Quality}_i + \beta_5 (\text{Post}_t \times \text{Barcelona}_c \times \text{Quality}_i) + \gamma X_i + \epsilon_{ict} \quad (2)$$

Table 3: OLS Regression Results

Variable	Coef.	Std. Err.	t	P< t	Lower Bound	Upper Bound
Intercept	-1306.2227	392.997	-3.324	0.001	-2076.936	-535.509
C(quality_type)[T.stars]	-52.1593	78.665	-0.663	0.507	-206.431	102.113
C(neighborhood)[T.Barajas, Madrid]	109.6326	312.007	0.351	0.725	-502.250	721.515
C(neighborhood)[T.Barcelona]	-89.9363	555.236	-0.162	0.871	-1178.820	998.947
C(neighborhood)[T.Carabanchel, Madrid]	-270.7705	381.241	-0.710	0.478	-1018.429	476.888
C(neighborhood)[T.Chamartín, Madrid]	-107.2322	271.936	-0.394	0.693	-640.530	426.066
C(neighborhood)[T.Ciutat Vella, Barcelona]	167.4667	137.170	1.221	0.222	-101.540	436.474
C(neighborhood)[T.Eixample, Barcelona]	323.2271	123.772	2.611	0.009	80.495	565.959
C(neighborhood)[T.Sants-Montjuïc, Barcelona]	1369.3217	175.660	7.795	0.000	1024.833	1713.811
Barcelona	165.4883	206.722	0.801	0.423	-239.918	570.895
Event	32.9616	79.240	0.416	0.677	-122.438	188.361
Barcelona:Event	979.8146	714.850	1.371	0.171	-422.090	2381.719
rating	120.9755	42.899	2.820	0.005	36.846	205.105
Barcelona:Event:rating	98.7716	86.103	1.147	0.251	-70.087	267.630
quality_over_5	374.9041	41.937	8.940	0.000	292.661	457.147
num_reviews	-0.0737	0.018	-4.174	0.000	-0.108	-0.039
max_tfidf_hotel	26.1867	11.548	2.268	0.023	3.539	48.834
unique_tfidf_word_cnt	9.1515	5.860	1.562	0.119	-2.340	20.643

Table 4: Regression Statistics

R-squared	0.328
Adj. R-squared	0.315
F-statistic	26.39
Prob (F-statistic)	2.45e-148
Observations	2098
AIC	36230
BIC	36450
Durbin-Watson	1.736

Regression for DID Model with Hotel Quality Impact

Estimated Difference-in-Differences (DID) Effect: 979.8146

Estimated DID Effect Modified by Hotel Rating: 98.7716

Note - Regression tables above shortened with some neighborhood coefficients removed. Refer to jupyter notebook MobileCongress_BCN_DID_Model.v2.ipynb for complete tables

Part 3c - Text features can be further developed to isolate impact of hotel quality on prices. The current result shows positive impact of quality, but analysis here might be biased because of multicollinearity and other issues.